

HealthPH: A Mobile and Web-Based Multilingual AI Platform for Respiratory Disease Surveillance Using Geospatial Social Media Analytics

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Abstract—The Philippines continues to face recurring respiratory disease outbreaks, compounded by limited healthcare access, fragmented surveillance systems, and significant linguistic diversity. Traditional monitoring approaches often lack real-time, multilingual capacity and geospatial granularity, creating blind spots in early detection and response. To address these challenges, this study presents HealthPH, a mobile and web-based multilingual AI platform for respiratory disease surveillance that integrates social media analytics, transformer-based natural language processing (NLP), and geospatial visualization. The system was developed using Rapid Application Development (RAD), with a backend powered by FastAPI and MongoDB, a ReactJS web dashboard, and a Progressive Web App (PWA) for mobile deployment. Core functionalities include multilingual classification of posts (English, Filipino, Cebuano) using fine-tuned mBERT and ELECTRA models, named entity recognition for symptom extraction, and geospatial mapping through OpenStreetMap APIs. Validation involved pilot deployment with local government units, focusing on usability, trend detection, and regional interpretability. Results demonstrate that HealthPH effectively identifies symptom mentions across multiple languages and generates real-time, location-specific analytics through interactive dashboards and mobile interfaces. By bridging gaps in linguistic inclusivity, real-time surveillance, and geospatial analysis, HealthPH contributes to advancing digital epidemiology in low-resource, multilingual contexts. Future work will expand language coverage, strengthen safeguards for data privacy and ethical use, and explore participatory reporting to enhance system responsiveness.

Keywords—Public Health Surveillance, Digital Epidemiology, Multilingual Natural Language Processing, Mobile and Web Application, Artificial Intelligence

I. INTRODUCTION

The Philippines faces persistent public health challenges, including recurrent respiratory disease outbreaks, uneven healthcare access, and limited surveillance infrastructure. In

2023, the Department of Health (DOH) reported over 150,000 cases of influenza-like illness (ILI), while the World Health Organization (WHO) estimated that the country accounted for 6.8% of the global tuberculosis (TB) burden, with approximately 739,000 cases annually and a treatment coverage rate below 70% [1]. The COVID-19 pandemic further exposed systemic inefficiencies, particularly the lack of real-time, multilingual, and community-level disease monitoring [2]. A 2024 national review revealed a physician density of only 7.92 per 10,000 population—below the WHO’s recommended threshold of 10, with nearly half of municipalities lacking a licensed physician and 43% of barangays without regular access to health workers [3]. In response to these constraints, digital platforms offer a promising avenue for enhancing public health surveillance[4]. Filipinos are among the most active social media users globally, frequently sharing symptoms, seeking medical advice, and voicing health concerns on platforms such as Facebook [5], [6]. These user-generated data streams—when processed using artificial intelligence (AI), machine learning (ML), and natural language processing (NLP) can provide real-time, community-level insights across the country’s linguistically diverse regions. This study introduces HealthPH, a multilingual, AI-powered surveillance platform designed to detect and monitor respiratory diseases using geospatial social media analytics. Unlike prior Philippine studies focused on sentiment analysis or thematic shifts, HealthPH offers a comprehensive architecture encompassing data acquisition, expert-informed annotation, transformer-based classification, named entity recognition, and user-facing interfaces. By integrating deep learning models with geospatial visualization and real-time analytics, HealthPH addresses critical gaps in conventional surveillance systems and supports inclusive, evidence-informed public health interventions in low-resource settings.

II. RELATED WORK

The rapid growth of social media has transformed platforms such as Facebook, Twitter, Reddit, and TikTok into valuable resources for public health surveillance. Wilson et al. [7] emphasized that these platforms generate millions of posts per minute, providing vast user-generated data that, when processed using NLP and machine learning, can detect and forecast outbreaks of diseases such as Zika, Dengue, MERS, and Ebola. Their study showed that social media often anticipated incidence trends weeks ahead of traditional systems, with statistically significant correlations between symptom-related keywords and epidemiological data, and that real-time mentions during COVID-19 improved prediction accuracy of infection rates. Extending this perspective, Terry et al. [8] reviewed 338 studies and highlighted three key functions of social media in health crises: surveillance, risk communication, and disease control. They noted an exponential rise in research post-COVID-19, yet also identified persistent gaps, including the dominance of studies in high-income, English-speaking countries and inconsistent integration into formal health systems[9]. In the Philippine context, Cruz et al. [10] found that Twitter reflected public sentiment during the 2019 elections, though online sentiments diverged from actual results due to sampling bias, with younger, urban, and technologically literate demographics overrepresented. Further, Cruz et al. [11] showed that social media topics evolved dynamically, shifting from policy discussions to misinformation and trust issues, suggesting that health-related discussions during outbreaks may similarly progress from symptom reporting to vaccine hesitancy or conspiracy theories. Wang et al. [12] provided a broader review of machine learning approaches, reporting that predictive models leveraging Twitter and Facebook data often outperformed traditional systems in speed and sensitivity. They also highlighted the success of transformer models like BERT in influenza prediction, but noted ongoing challenges in filtering noise, handling sarcasm, and addressing the overemphasis on English datasets, which limits generalizability to low-resource languages. Within the Philippines, Delizo et al. [13] demonstrated the utility of Twitter for COVID-19 surveillance, achieving 72.17% accuracy and showing sentiment shifts aligned with key government interventions. Tobias et al. [5] mapped global literature, identifying machine learning, NLP, and sentiment analysis as dominant methods, with Twitter as the most common source, and noted the limited yet growing contributions from Southeast Asia. More recently, Maceda et al. [14] highlighted the promise of large language models, showing that GPT-4 outperformed fine-tuned mBERT in classifying sentiments in low-resourced, code-mixed Philippine languages, though they cautioned about prompt sensitivity and challenges in noisy environments. Traditional classifiers such as k-NN, SVMs, and logistic regression provided early baselines, but have been surpassed by transformer models like mBERT and XLM-RoBERTa. Dufter and Schütze [15] showed that mBERT achieved higher multilingual generalization than traditional TF-IDF models, while Libovický et al. [16] confirmed superior cross-lingual alignment, enhanced further by centering techniques. Gonen et al. [17] demonstrated that mBERT encodes word-level translations even without explicit supervision, outperforming static embeddings and enabling zero-shot transfer across languages. Despite these strengths,

challenges remain for low-resource languages. Muller et al. [18] revealed that cross-lingual transfer depends heavily on lower-layer multilingual encoders, and that reinitialization significantly reduced parsing and NER performance. Philippine languages further complicate this due to agglutinative morphology, code-mixing, and borrowing from Spanish and English. Pilar et al. [19] illustrated these difficulties through the CEBUANER model, which performed well in Cebuano but degraded sharply when transferred to Tagalog, underscoring the limits of cross-lingual generalization even among related languages.

III. METHODS

A. System Deployment

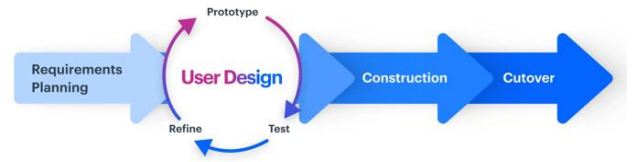


Fig. 1. Rapid Application Development

Rapid Application Development (RAD) methodology was implemented to accelerate the delivery of a functional and user-centered digital health surveillance system, as reflected in Fig. 1.

1) System Architecture

The HealthPH Toolkit comprises two primary components: a web-based dashboard for centralized data analysis and a mobile application for field-level access.

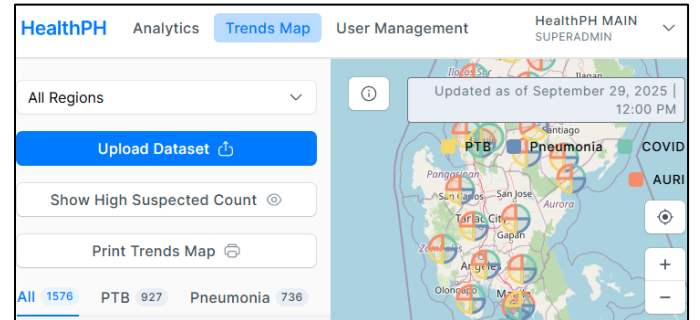


Fig. 2. HealthPH Disease Surveillance Web Dashboard

a) Backend Architecture

The backend was developed using FastAPI, a high-performance web framework for asynchronous API operations. It interfaces with a MongoDB database that stores annotated posts, user credentials, model outputs, and geospatial metadata. Data pipelines include preprocessing modules, model inference routines, and a geolocation handler using OpenStreetMap APIs.

An AI inference layer integrates the fine-tuned mBERT and ELECTRA models for real-time annotation of new data uploads. Each post is classified into one or more disease categories, and detected symptom entities are logged with timestamp and location metadata. Authentication is managed via OAuth2 protocols, and role-based access control (RBAC) is implemented to distinguish between administrators, data annotators, and public health officers.

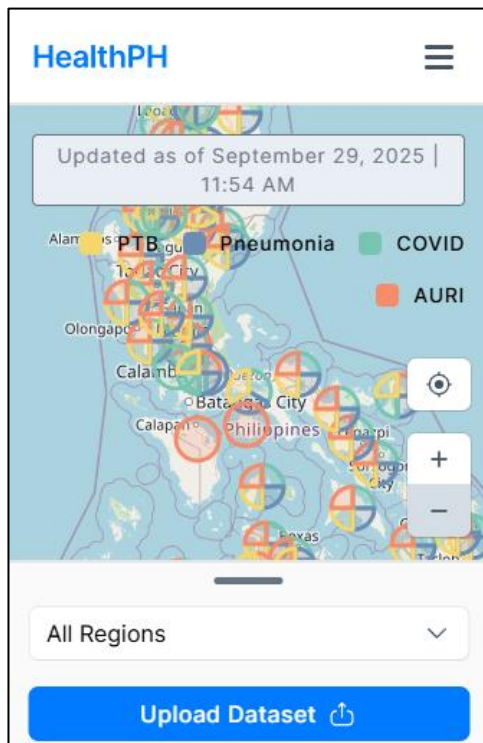


Fig. 3. HealthPH A popup that shows extracted symptoms when a marker has been tapped / pressed.

2) Web Dashboard

The web dashboard was implemented using ReactJS. Its core functionalities include:

a) Trends Map Module.

A dynamic, interactive map displaying spatially distributed symptoms by disease category. Users can filter by region or condition and view post-level details via markers and pop-ups.

b) Analytics Module

A collection of visual analytics, including symptom frequency histograms, word clouds, and disease-specific pie charts. These are dynamically updated based on dataset uploads.

c) Data Upload and Management

Users can upload annotated datasets and generate downloadable reports (PDF) with map snapshots and analytic summaries. Security features include two-factor authentication, activity logging, and restricted access to upload and administrative functions. Figure 2 reflects the HealthPH Disease Surveillance Web Dashboard.

3) Mobile Application

The mobile application was built as an Android-based Progressive Web App (PWA) using the same API endpoints as the web dashboard. It offers real-time views of regional disease trends, interactive map features, and analytics charts, as shown in Fig. 3. Access is authenticated, and functionality is optimized for viewing rather than uploading or modifying datasets. The application supports offline caching and lightweight rendering for rural deployment.

B. Validation

Following model evaluation, the classifiers were integrated into a backend inference engine. The system was subjected to a pilot deployment in collaboration with local government units (LGUs) and stakeholders. Stakeholder feedback was incorporated into the interface design and analytics functionality. Validation focused on usability, regional data interpretation, and response applicability.

IV. RESULTS

This study presents the development and validation of the HealthPH Toolkit, a multilingual, AI-powered public health surveillance system designed for early detection of respiratory disease symptoms from user-generated social media content. The system comprises a web-based dashboard and a mobile application, both engineered to support real-time trend analysis, spatial visualization, and epidemiological reporting. Its architecture integrates transformer-based language models, a custom-curated dataset, and geospatial analytics modules, addressing a documented gap in scalable disease surveillance for linguistically diverse and resource-constrained settings. The machine learning pipeline implemented in the HealthPH Toolkit utilized both traditional and transformer-based multilabel classification models. The classification task involved identifying four disease categories: Acute Upper Respiratory Infection (AURI), Pneumonia, Pulmonary Tuberculosis (PTB), and COVID-19. Model training was conducted using the RespiratoryPH dataset [20].

The HealthPH web dashboard was developed using ReactJS, with FastAPI used for API orchestration and MongoDB for data storage. The system supports dataset ingestion, trend visualization, and analytical reporting. Its interactive Trends Map module overlays extracted symptom clusters on geospatial data using OpenStreetMap, enabling location-specific analysis of suspected disease cases. A complementary Analytics module provides descriptive visualizations such as frequency distributions, pie charts, and word clouds. These features enhance the interpretability for health officials, allowing for evidence-informed public health responses. The mobile application, built as a Progressive Web App (PWA), offers similar data views with a focus on field accessibility and supports secure authentication and user-level permissions.

In evaluating the broader impact of the HealthPH Toolkit, it is evident that its design aligns with multiple targets of the United Nations Sustainable Development Goals (SDGs). The system contributes directly to SDG 3 (Good Health and Well-being) by enhancing real-time surveillance capabilities for communicable diseases. Its linguistic inclusivity supports SDG 10 (Reduced Inequalities) by incorporating non-dominant languages into national-level data analytics. The toolkit's regional deployment and integration with local government units further contribute to SDG 11 (Sustainable Cities and Communities) by enabling localized epidemic preparedness. Educational and research partnerships developed through the project promote SDG 4 (Quality Education) and SDG 9 (Industry, Innovation, and Infrastructure), while the cross-sectoral nature of implementation, involving public agencies, academic institutions, and community stakeholders, reflects the principles of SDG 17 (Partnerships for the Goals).

V. CONCLUSIONS AND FUTURE WORK

The HealthPH Toolkit demonstrates how multilingual NLP, geospatial analytics, and real-time visualization can be integrated into a scalable system for respiratory disease surveillance in the Philippines. By extending coverage beyond English to Filipino and Cebuano, the platform addresses critical gaps of linguistic exclusion and geographic blind spots that limit conventional surveillance. Future work will prioritize expanding linguistic coverage, incorporating participatory feedback from local stakeholders, and fine-tuning domain-specific language models to improve accuracy and robustness. Together, these efforts will advance HealthPH as an inclusive, reliable, and context-sensitive digital epidemiology platform with relevance for other low-resource, multilingual settings.

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